

4.3 Basic Operations on Data using Pandas

پینڈاژ ٹیبل نما ڈیٹا فریم دیتا ہے۔

CS XII

Chapter 4: Data and Analysis

Lecture 04 of 8

★ Golden

4.3 Basic Operations on Data using Pandas ★ Golden

LEARN IT

Pandas is a third-party library for tables. Install once: `pip install pandas` in the terminal (not inside the `.py` file). Convention: `import pandas as pd`.

A **DataFrame** is a 2-D table of rows and columns, like Excel or an SQL table. You can filter, sort and calculate whole columns without writing a `for` loop.

```
import pandas as pd
data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Math": [85, 90, 78],
    "Science": [80, 88, 70],
}
df = pd.DataFrame(data)
print(df)
```

OUTPUT

	Name	Math	Science
0	Ali	85	80
1	Sara	90	88
2	Ahmed	78	70

`read_csv('students.csv')` and `read_excel('students.xlsx', sheet_name='Sheet1')` load files. `df.head()` prints the first five rows — use it on large files so you do not flood the screen.

```
df["Total"] = df["Math"] + df["Science"]
df["Average"] = df["Total"] / 2
print(df)
```

OUTPUT

	Name	Math	Science	Total	Average
0	Ali	85	80	165	82.5
1	Sara	90	88	178	89.0
2	Ahmed	78	70	148	74.0

WORKED EXAMPLE

Check: Ali $85+80=165$, $165/2=82.5$. Sara $90+88=178$, 89.0 . Ahmed $78+70=148$, 74.0 .

اردو میں سمجھیں

پینڈا ز ٹیبل نما ڈیٹا فریم دیتا ہے۔ ڈکشنری، سی ایس وی یا ایکسل سے لوڈ کریں۔ نئے کالم بغیر لوپ کے پوری قطار پر حساب کر کے بنتے ہیں۔ `head()` صرف پہلی پانچ قطاریں دکھاتا ہے۔

CHECK YOURSELF

1. What is `pd`?
2. Why not `print(df)` on a million-row file?

Answers: 1. The usual alias for pandas. 2. It would print everything; use `df.head()`.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Common DS operations include:

- a. Insert, delete, search, traverse
- b. Only compile
- c. Only ping
- d. Only print the logo

MCQ 2. Traverse means:

- a. Visit elements in some order
- b. Delete all
- c. Encrypt all
- d. Sort the OS

MCQ 3. Search cost depends on:

- a. The structure (array vs sorted vs hash vs tree)
- b. The PDF theme
- c. The teacher name
- d. The sprint

MCQ 4. Insert in a full array may:

- a. Fail or require resizing
- b. Always $O(1)$ no matter what
- c. Create a graph edge automatically
- d. Pop a stack

MCQ 5. Delete from a queue happens at the:

- a. Front
- b. Rear only always
- c. Random index
- d. Root of a tree

MCQ 6. You choose a DS by the:

- a. Operations you will do most
- b. First letter of the course
- c. Colour
- d. NAND count

Extra short questions

1. List five operations.
2. Give costs qualitatively: array index vs list search.
3. What does traverse a tree mean?
4. Why "update" might be search+insert.

Extra long question

1. For array, stack, queue, tree: name the natural insert/delete ends and one costly operation. Table.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Create, insert, delete, search, traverse, update, sort (as applicable).
2. Array index $O(1)$; unsorted list search $O(n)$.
3. Visit every node (pre/in/post or level order).
4. You must find the item first.

Long

1. Array: insert middle costly. Stack: top only cheap. Queue: rear in front out. Tree: search/insert depend on balance. 8-row table.

Next lecture: 4.3.3 Handling Missing Values (NaN)